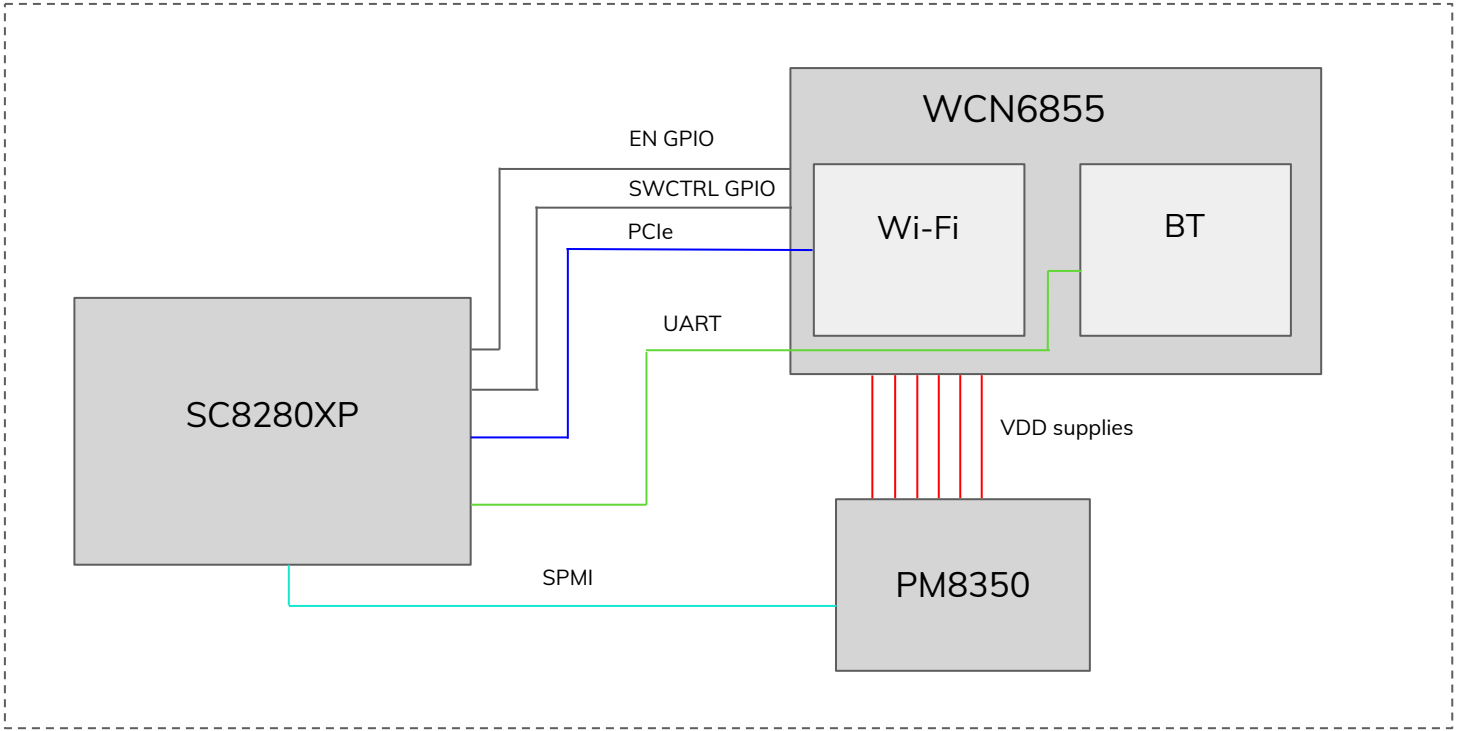
BoF: Powering up “discoverable bus-attached” devices on DT-based platforms



Agenda

- Devices attached to discoverable buses need to be powered up for discovery
- No generic way in kernel to do that currently
- USB Onboard Hub handles that by using a separate (fake?) platform device
- Proposed solution: introduce a PCI slot driver to power up the device
  - Problems:
    - Power sequence shared between multiple devices
    - Power sequence might be complex
  - Power Sequencing Subsystem ?
    - Could handle shared resources for both devices
    - Devicetree wise might not be HW accurate
  -

X13s: Wi-fi & Bluetooth



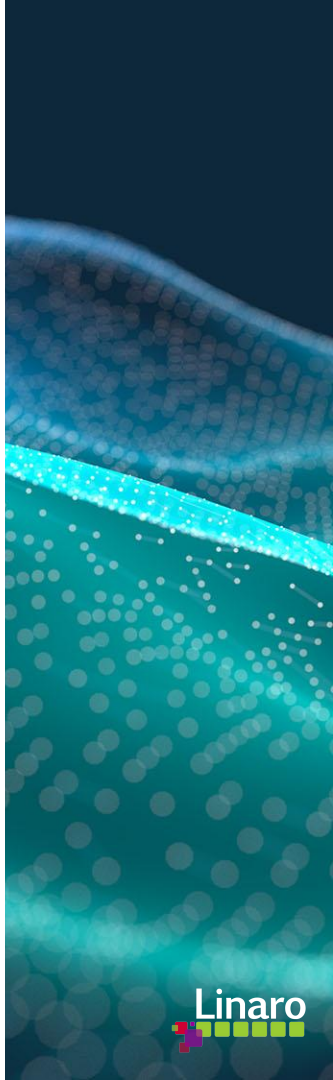
X13s: Bluetooth devicetree node

```
&uart2 {
        status = "okay";

        bluetooth {
                compatible = "qcom,wcn6855-bt";

                vddio-supply = <&vreg_s10b>;
                vddbtcxmx-supply = <&vreg_s12b>;
                vddrfacmn-supply = <&vreg_s12b>;
                vddrfa0p8-supply = <&vreg_s12b>;
                vddrfa1p2-supply = <&vreg_s11b>;
                vddrfa1p7-supply = <&vreg_s1c>;

                enable-gpios = <&tlmm 133 GPIO_ACTIVE_HIGH>;
                swctrl-gpios = <&tlmm 132 GPIO_ACTIVE_HIGH>;
              ,,,
        };
};
```



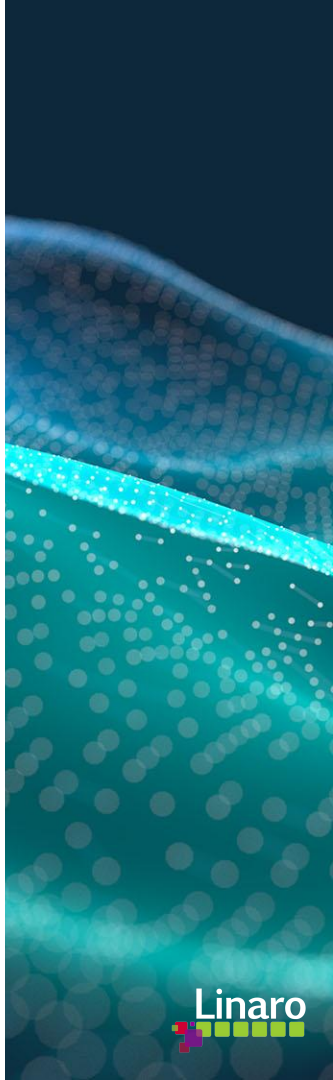
X13s: Wi-fi devicetree node

```
&pcie4 {
	...

	pcie@0 {
		device_type = "pci";
		reg = <0x0 0x0 0x0 0x0 0x0>;

		bus-range = <0x01 0xff>;

		wifi@0 {
			compatible = "pci17cb,1103";
			reg = <0x10000 0x0 0x0 0x0 0x0>;
		};
	};
};
```



Thank you

